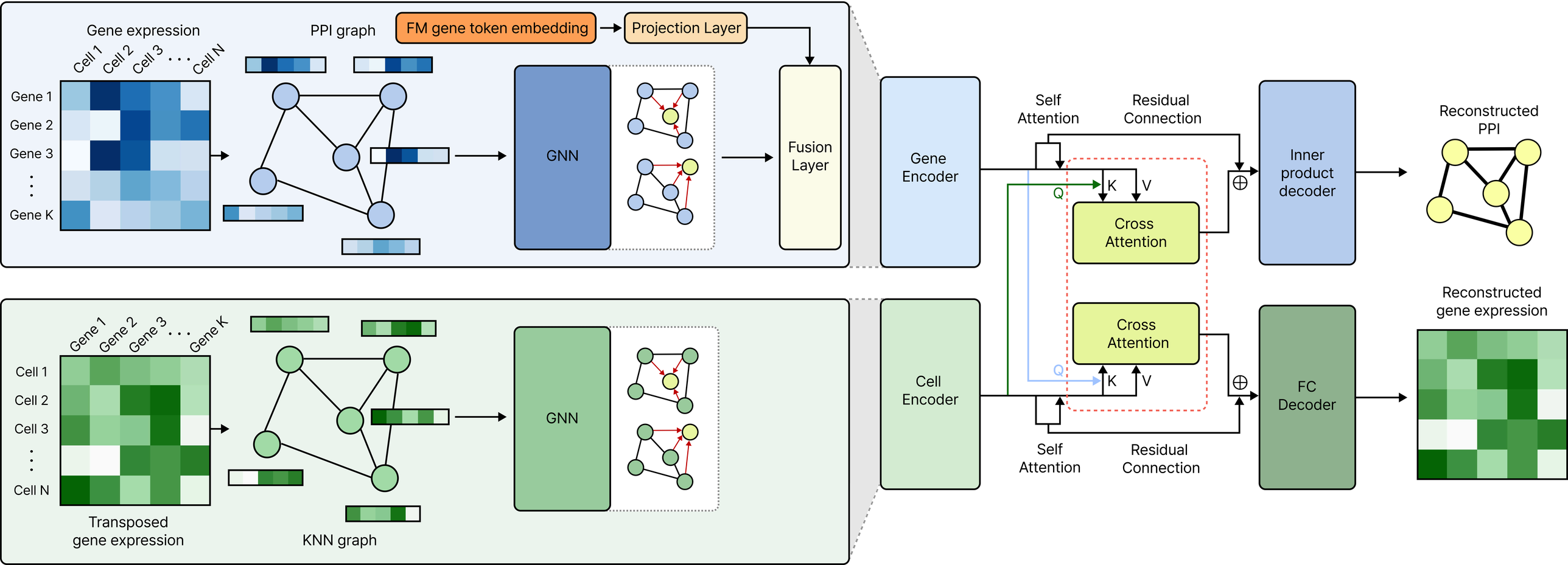
Gene expression
Cell 1
Cell 2
Cell 3
Cell N
Gene 1
Gene 2
Gene 3
Gene K
PPI graph
FM gene token embedding
Projection Layer
GNN
Fusion Layer
Gene Encoder
Self Attention
Residual Connection
Q
K
V
Cross Attention
Inner product decoder
Reconstructed PPI
Gene 1
Gene 2
Gene 3
Gene K
Cell 1
Cell 2
Cell 3
Cell N
Transposed gene expression
KNN graph
GNN
Cell Encoder
Cross Attention
Self Attention
Residual Connection
FC Decoder
Reconstructed gene expression